

Programme	B. Sc. Computer Science				
Course Title	Advanced Distributed Computing				
Type of Course	Elective				
Semester	VIII				
Academic Level	400 - 499				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	4	-	-	60
Pre-requisites	Basic knowledge in data structures and operating systems.				
Course Summary	The syllabus is prepared with the view of preparing the BSc Computer Science Graduates to understand the system models, algorithms and protocols that allow computers to communicate and coordinate their actions to solve a problem. This course helps the learner to understand the distributed computation model and various concepts like global state, termination detection, mutual exclusion, deadlock detection, shared memory				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Summarize various aspects of distributed computation model	U	C	Instructor-created exams / Quiz
CO2	Appreciate and apply Distributed Computing and Communication design principles	U	C	Instructor-created exams / Quiz
CO3	Illustrate election algorithm, global snapshot algorithm and termination detection algorithm.	U	P	Instructor-created exams / Quiz
CO4	Compare token based, non-token based and quorum based mutual exclusion algorithms.	U	C	Instructor-created exams / Case studies
CO5	Recognize the significance of deadlock detection and shared memory in distributed systems	U	P	Instructor-created exams / Quiz Case studies
CO6	Understand the concepts of failure recovery and consensus	U	P	Instructor-created exams / Quiz /Case studies
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C)# - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs	Marks
I	Introduction to Distributed Systems		12	15
	1	Introduction to Distributed Systems: Goals of the Distributed Systems, Relation to parallel systems	2	
	2	Synchronous versus asynchronous execution, Design issues and challenges	2	
	3	Types of Distributed Systems	2	
	4	Distributed System Models	3	
	5	Hardware and software concepts related to distributed systems, middleware models .	3	
II	Distributed Computing and Communication		12	15
	6	Distributed Computing and Communication design principles: A Model of distributed executions	2	
	7	Models of communication networks, Global state of distributed system,	1	
	8	Models of process communication.	3	
	9	Communication and Coordination: Shared Memory, Consistency, Atomicity	3	
	10	Message- Passing, Consensus, Conditional Actions, Critical Paths	1	
	11	Scalability, and cache coherence in multiprocessor systems, synchronization mechanism.	2	
III	Election algorithm, Global state and Termination detection		12	20
	12	Logical time – A framework for a system of logical clocks, Scalar time, Vector time	2	
	13	Leader election algorithm – Bully algorithm, Ring algorithm	2	
	14	System model and definitions, Snapshot algorithm for FIFO channels – Chandy Lamport algorithm.	2	
	15	Termination detection – System model of a distributed computation,	2	
	16	Termination detection using distributed snapshots,	2	
	17	Termination detection by weight throwing ,Spanning-tree-based algorithm	2	
IV	Mutual exclusion and Deadlock detection		12	20
	18	Distributed mutual exclusion algorithms – System model, Requirements of mutual exclusion algorithm	2	
	19	Quorum-based mutual exclusion algorithms – Maekawa's algorithm	2	
	20	Token-based algorithm – Suzuki-Kasami's broadcast algorithm.	3	
	21	Deadlock detection in distributed systems – System model, Deadlock handling strategies	2	
	22	Issues in deadlock detection, Models of deadlocks	3	

V	Open Ended Module	12	
	CASE STUDY: Distributed shared memory and Failure recovery Lamport's bakery algorithm. Check pointing and rollback recovery – System model, consistent and inconsistent states, Different types of messages, Issues in failure recovery, Checkpoint based recovery, Log based roll back recovery.		

Reference Books:

1. Ajay D. Kshemkalyani and Mukesh Singhal, Distributed Computing: Principles, Algorithms, and Systems, Cambridge University Press, 2011
2. George Coulouris, Jean Dollimore, Tim Kindberg and Gordon Blair. Distributed Systems: Concepts and Design, Addison Wesley, Fifth edition.
3. Kai Hwang, Geoffrey C Fox, Jack J Dongarra, Distributed and Cloud Computing – From Parallel Processing to the Internet of Things, Morgan Kaufmann Publishers, 2012.
4. Sukumar Ghosh, Distributed Systems: An Algorithmic Approach, CRC Press, Second edition, 2015.
5. Maarten Van Steen, Andrew S. Tanenbaum, Distributed Systems, Prentice Hall of India, Third edition, 2017.
6. Randy Chow and Theodore Johnson, Distributed Operating Systems and Algorithm Analysis, Pearson Education India, First edition, 2009.
7. Valmir C. Barbosa, An Introduction to Distributed Algorithms, MIT Press, 2003.

Mapping of COs with PSOs and POs :

	PSO1	PSO2	PSO3	PSO4	PSO5	PSO6	PO1	PO2	PO3	PO4	PO5	PO6
CO1	-	2	-	-	-	-						
CO2	-	3	-	-	-	-						
CO3	-	2	1	-	1	1						
CO4	-	2	1	2	1	2						
CO5	-	3	1	-	-	2						
CO6	-	2	-	-	-	2						

Correlation Levels:

Level	Correlation
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-	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Quiz / Assignment/ Quiz/ Discussion / Seminar
- Midterm Exam
- Assignments (20%)
- Final Exam (70%)

Mapping of COs to Assessment Rubrics :

	Internal Exam	Assignment	Project Evaluation	End Semester Examinations
CO 1	✓			✓
CO 2	✓			✓
CO 3	✓	✓		✓
CO 4		✓		✓
CO 5		✓		✓
CO 6			✓	